

VYSHNAV K KUMAR

FULL STACK DEVELOPER | AWS CLOUD-NATIVE | MERN | GENERATIVE AI & RAG

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years building and scaling enterprise web applications across React.js, Next.js, Node.js, Express.js, and MongoDB, with hands-on cloud deployment experience on AWS EC2, Docker, and Kubernetes fundamentals. Track record of designing REST/GraphQL APIs, microservices, and real-time systems, and of maintaining 99.9% service availability through CI/CD automation and infrastructure hardening. Extends full-stack delivery into Generative AI application development — LLM API integration, prompt engineering, retrieval-augmented generation (RAG), embeddings, and vector search. Proven technical lead guiding a 9-member engineering team, improving application performance by 40%, and cutting environment setup time by 83%.

TECHNICAL SKILLS

Cloud & DevOps	AWS (EC2), Docker, Kubernetes fundamentals, CI/CD, GitLab CI/CD, Nginx, Linux, Git, Infrastructure Monitoring
Backend & APIs	Node.js, Express.js, REST API Development, GraphQL, WebSocket, JWT, Authentication & Authorization, Microservices
Frontend	React.js, Next.js, Redux, Redux Toolkit, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Material UI, Responsive Design
Databases & Caching	MongoDB, MySQL, PostgreSQL, Mongoose, Sequelize, Redis
Generative AI	LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, OpenAI API concepts, Gemini API concepts, LangChain/LlamaIndex concepts, Agentic AI
Engineering Practice	System Design, API Integration, Agile/Scrum, Code Review, TDD, Performance Optimization, Postman, Jira

PROFESSIONAL EXPERIENCE

Technical Lead | Infinite Open Source Solutions LLP

Calicut, India | Jan 2024 – Present

- Lead full-stack architecture and delivery of enterprise MERN applications, deployed via Dockerized, AWS EC2-based environments with CI/CD automation.
- Maintained 99.9% service availability through Dockerized deployments, CI/CD pipelines, monitoring, infrastructure hardening, and incident-response practices.
- Engineered an automated demo provisioning pipeline that reduced setup time by 83% (90 seconds to 15 seconds), saving 10+ engineering hours per week.
- Improved application performance by up to 40% through query optimization, API profiling, caching, and systematic root-cause analysis.
- Design scalable backend APIs, authentication flows, real-time features, and responsive UIs, translating product requirements into maintainable technical solutions.
- Guide a cross-functional team of 9 engineers through sprint planning, code reviews, technical decisions, debugging, and release management.
- Contribute to AI-enabled solution design involving LLM-based summarization, structured outputs, document processing, embeddings, vector retrieval, and RAG-oriented workflows.
- Integrated AI-assisted development workflows using Claude Code, Cursor, GitHub Copilot, and ChatGPT for rapid prototyping, refactoring, debugging, and test generation.

- Developed end-to-end product features using React.js, Node.js, Express.js, MongoDB, MySQL, and RESTful APIs for enterprise web platforms.
- Modernized legacy modules into service-oriented and microservices-based components, improving scalability, maintainability, and deployment flexibility.
- Implemented real-time dashboards and event-driven functionality using WebSocket, Redis, and RabbitMQ-based processing patterns.
- Built reusable UI components, modular backend services, validation layers, and secure JWT-based authentication.
- Resolved 50+ critical production issues through log analysis, performance profiling, database investigation, and regression testing.

SELECTED PROJECTS

AI-Powered Customer Engagement Platform

React.js Node.js Python/FastAPI MongoDB PostgreSQL

LLMs / RAG

- Designed solution components for an AI-enabled platform generating summaries, requirements, searchable knowledge, and sentiment insights from conversations and documents, using LLM integration, prompt workflows, embeddings, vector retrieval, RAG, background queues, and multi-tenant API design.

Learning Management System

MERN Stack AWS WebSocket

- Developed and deployed on AWS an LMS platform with responsive interfaces, course/user management, progress tracking, automated assessments, real-time functionality, and analytics dashboards.

Process Automation & DevOps Pipeline

Node.js Docker GitLab CI/CD AWS

- Created automated provisioning and deployment workflows on AWS that reduced setup time by 83%, improved release consistency, and supported 99.9% service availability across production environments.

Network Marketing Enterprise Platform

React.js Node.js Express.js MongoDB MySQL Redis

- Built and scaled a multi-tenant platform with hierarchical user structures, commission engines, financial workflows, genealogy dashboards, and high-volume API/database operations.

EDUCATION & CERTIFICATIONS

B.Tech in Electronics and Communication Engineering — T K M College of Engineering

Kollam, Kerala

Certifications: Penetration Tester (RedTeam Hacker Academy) | Full Stack Development (CCBP Tech 4.0 Intensive)

ACHIEVEMENTS

- **Best R&D Performer Award:** Recognized for product innovation, backend optimization, and key engineering improvements.
- **Process Optimization Excellence:** Honored for reducing custom demo setup time by 83% and significantly boosting sales enablement.